uc3m | Universidad Carlos III de Madrid

Master on Statistics for Data Science

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Multivariate Analysis

MDS and Cluster

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1. MDS

Multidimensional Scaling (MDS) is a classical multivariate technique used to describe and visualize the structure of a dataset by representing units in a low-dimensional Euclidean space. Instead of starting from raw quantitative variables (as PCA does), MDS begins with a distance matrix, that is, a symmetric matrix of pairwise distances between units, where each entry quantifies how statistically different two units are from each other: small values indicate very similar units, larger values indicate more distinct units, and a value of zero usually corresponds to identical units. Formally, a distance (or quasi-metric) $\delta : \mathcal{E} \times \mathcal{E} \to \mathbb{R}$ satisfies three basic properties: $\delta_{ij} \geq 0$ for all units i, j, $\delta_{ii} = 0$ for every unit i, and symmetry $\delta_{ij} = \delta_{ji}$. The entries of the distance matrix may come from a true metric (such as the Euclidean distance for quantitative variables) or from more general separation measures adapted to the nature of the data, such as the Gower distance for mixed data or specific coefficients for binary or categorical variables. Once this distance matrix is defined, MDS searches for a set of orthogonal axes, called principal coordinates, on which Euclidean distances reproduce, as closely as possible, the distances provided as input, thereby yielding a low-dimensional configuration of the units in which their relative positions can be directly interpreted and visualized. The main advantage of MDS is that it relies solely on the dissimilarity structure of the data encoded in the distance matrix. Consequently, it is applicable to datasets comprising variables of any type, including mixed datasets, provided that an appropriate distance (or dissimilarity) function is specified for the data being analyzed, whether quantitative, binary, qualitative, or mixed.

1.1. Principal Components Search

The data used in this study comes from the *Encuesta sobre el Gasto de los Hogares en Educación (EGHE) for the academic year 2023/2024*, published by the Spanish National Statistics Institute (INE). After the data cleaning and preprocessing steps described in the previous assignment, let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote the final data matrix, collecting the variables used in this study, where $n = 374$ is the number of individuals and $p = 10$ is the number of variables. Specifically, the dataset contains six numerical variables: 'age' (X_1), 'uniform' (X_2), 'food' (X_3), 'books' (X_4), 'computer' (X_5), and 'activities' (X_6); two binary variables: 'sex' (X_7) and 'help' (X_8); and two categorical variables: 'nationality' (X_9) and 'type' of educational institution (X_{10}). As detailed in the previous work, the numerical variables have already been preprocessed to mitigate strong right-skewness and differences in scale. In particular, a logarithmic transformation was applied to all numerical variables except 'food' (X_3), and, subsequently, all numerical variables were centred and standardized. Thus, the matrix $\mathbf{X}$ used in the present analysis contains the log-transformed (where appropriate) and standardized numerical variables, together with the original binary and categorical variables.

Since we are working with mixed-type data, we use the Gower similarity to compute the matrix of squared distances. In order to apply Gower's measure, the columns of the data matrix must be arranged in the following order: quantitative variables, binary variables, and categorical variables, and we must also specify how many variables of each type are included.

Let p_1, p_2, and p_3 denote, respectively, the numbers of quantitative, binary, and categorical variables in $\mathbf{X}$, so that $p_1 + p_2 + p_3 = p$. For each pair of individuals (i, j), Gower's similarity coefficient s_{ij} is defined as

$$s_{ij} = \frac{\displaystyle\sum_{h=1}^{p_1}\Big(1 - |x_{ih} - x_{jh}|/G_h\Big) + a + \alpha_{ij}}{p_1 + (p_2 - d) + p_3}, \qquad 0 \leq s_{ij} \leq 1 \tag{1.1}$$

where

- For the p_1 quantitative variables, Gower's contribution is based on an adapted Manhattan distance: the absolute difference $|x_{ih} - x_{jh}|$ (Manhattan on a single coordinate) is divided by the range G_h of the h-th variable, so that the similarity term $1 - |x_{ih} - x_{jh}|/G_h$ lies in $[0, 1]$, equals 1 when the two observations coincide, and decreases linearly as they differ.

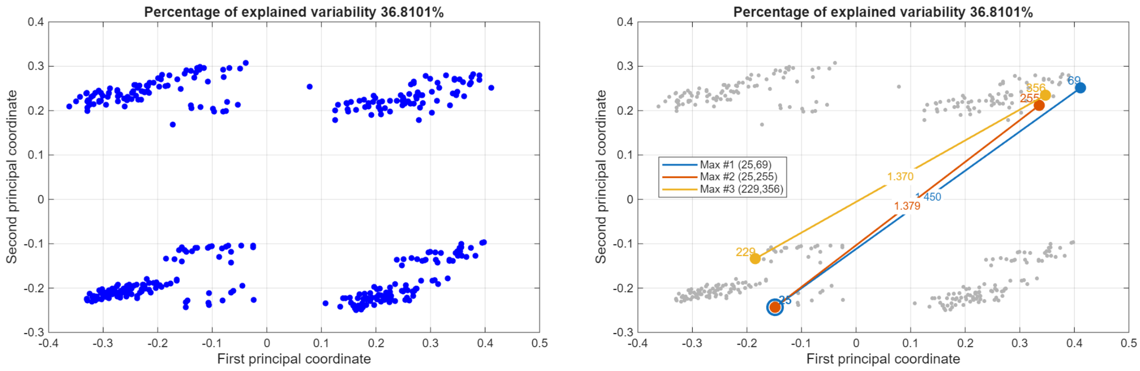
- For the p_2 binary variables,Jaccard-type similarity is used: a is the number of $(1, 1)$ matches and d the number of $(0, 0)$ matches among the asymmetric binary variables. Only joint presences contribute to similarity, since $(1, 1)$ matches carry information, whereas $(0, 0)$ matches usually do not.

- For the p_3 categorical variables, the matching coefficient is employed. Here α_{ij} counts the number of variables for which both observations fall in the same category. Each categorical variable contributes 1 if $x_{ik} = x_{jk}$ and 0 otherwise.

Using the MATLAB function `gower2`, we compute the similarities (1.1) for all pairs of individuals (i, j). From these, we define the matrix of squared distances $\mathbf{D}^{(2)} = (\delta^2_{ij})_{1\leq i,j\leq n}$ by setting $\delta^2_{ij} = 1 - s_{ij}$, with $0 \leq \delta^2_{ij} \leq 1$. This $n \times n$ matrix $\mathbf{D}^{(2)}$ is the input for classical MDS: given $\mathbf{D}^{(2)}$, we seek a configuration of the n units in orthogonal axes (an MDS configuration) such that the ℓ^2 pairwise distances between units reproduce, as closely as possible, the entries of $\mathbf{D}^{(2)}$.

Classical MDS proceeds by converting $\mathbf{D}^{(2)}$ into a Gram matrix $\mathbf{G} = -\frac{1}{2}\,\mathbf{H}\,\mathbf{D}^{(2)}\,\mathbf{H}$, where $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ is the centering matrix. Before proceeding, we must check whether the distance matrix is Euclidean, which holds if and only if the corresponding Gram matrix $\mathbf{G}$ is positive semidefinite. Then, assuming that $\mathbf{G}$ is positive semidefinite, it admits the spectral decomposition $\mathbf{G} = \mathbf{U}\,\mathbf{\Delta}\,\mathbf{U}^\top = \mathbf{Y}\,\mathbf{Y}^\top$, where $\mathbf{U}$ is an orthogonal matrix whose columns are the eigenvectors of $\mathbf{G}$ and $\mathbf{\Delta} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ is a diagonal matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. The matrix of principal coordinates is then defined as $\mathbf{Y} = \mathbf{U}\,\mathbf{\Delta}^{1/2}$, so that the k-th column of $\mathbf{Y}$ gives the coordinates of the individuals along the k-th principal axis, and the associated eigenvalue λ_k measures the amount of variation in the (squared) distances accounted for by that axis. The eigenvalues are ordered from largest to smallest, so the first axis explains the largest proportion of variability.

This procedure is implemented in the `coorp` function, which, after checking that $\mathbf{D}$ fulfills the Euclidean property, computes $\mathbf{G}$, inspects its eigenvalues, and returns the corresponding principal coordinates.



(a) MDS map (all points) (b) MDS map (top-3 distances)

Figure. 1.1. Multidimensional scaling maps based on Gower distance.

The MDS maps in Figure 1.1 depict the proportion of variability explained by the first two principal coordinates (36.8011 %) and the configuration of the individuals in the orthogonal plane formed by the first and second principal coordinates. The figures contain 374 points, corresponding to the 374 observations in the sample. The distances between points in this two-dimensional representation provide an approximate reproduction of the pairwise square distances between individuals given by the Gower distance matrix.

In particular, Figure 1.1b focuses on the same MDS configuration and highlights the three largest squared pairwise distances. The corresponding pairs of individuals are connected by colored line segments in the map, and each segment is annotated with the value of the squared distance. Take, for instance, the squared distance between individuals 25 and 69 in the MDS representation, $\delta^2_{25,69} = 1{,}4496$, given by the entry $(i, j) = (25, 69)$ of the Gower squared distance matrix $\mathbf{D}^{(2)}$. If we compute the squared Euclidean distance between individuals in the new coordinate system, $\delta^2_{25,69} = (y_{25} - y_{69})'(y_{25} - y_{69})$, we obtain the same value. Therefore, for this pair of individuals, the objective of MDS is satisfied: the distance in the map reproduces the squared distance given by the Gower distance matrix.

The plot in Figure 1.2 shows the cumulative percentage of total variability explained by the coordinates, computed from the eigenvalues of the inner-product matrix $\mathbf{G}$. As we can see, the first 10 coordinates account for about 75 % of the variability, and we do not reach 90 % until we include 27 coordinates. In addition, its geometric variability is 0,5872, indicating that on average, pairs of individuals differ by about 59 % of the maximum possible dissimilarity, so the dataset is fairly diverse rather than composed of very similar profiles.

```
disp(accum(1:10)')
20.9980
36.8101
48.5736
54.6213
58.7998
62.7275
66.1736
69.3607
72.2450
74.4691
```

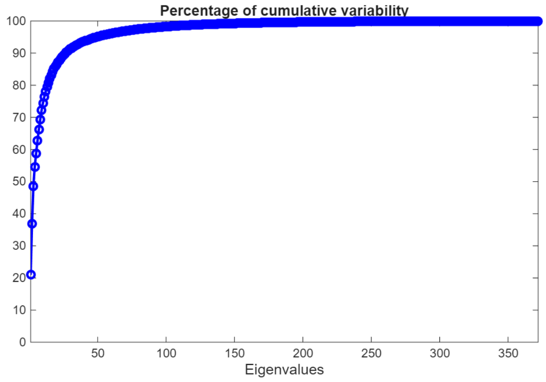
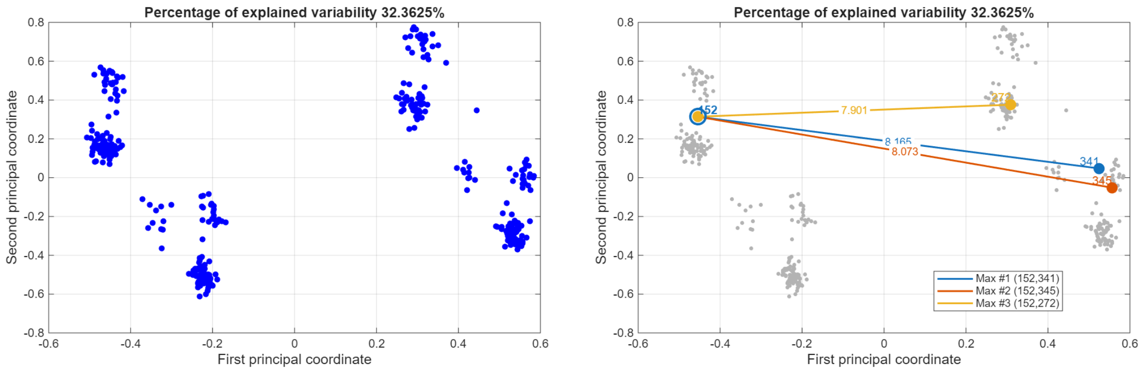


Figure. 1.2. Percentage of cumulative variability based on Gower distance.

Related Metric Scaling (RelMS) is known to provide a more robust and stable treatment of mixed-type variables than the classical Gower metric. Therefore, we now repeat the previous analysis using the RelMS-based distance matrix in place of Gower's distance. Starting from the pairwise RelMS dissimilarities, we construct the corresponding squared-distance matrix, obtain the associated Gram matrix, and compute the MDS configuration of principal coordinates. This allows us to compare the maps derived from RelMS with those based on Gower's metric and to assess any gain in robustness and stability in the representation of the data.

The multidimensional scaling maps based on the RelMS distance are displayed in Figure 1.4. The first two principal coordinates now explain 32.3625 % of the total variability, slightly less than in the Gower case (36.8101 %), but the overall cluster structure of the 374 individuals is preserved. As before, the Euclidean distances between points in this two-dimensional representation provide an approximate reproduction of the squared pairwise RelMS distances.



(a) MDS map (all points) (b) MDS map (top-3 distances)

Figure. 1.3. Multidimensional scaling maps based on RelMS distance.

Figure 1.3b highlights the three largest squared RelMS distances, corresponding to the pairs of individuals $(152, 341)$, $(152, 345)$ and $(152, 272)$, which are joined by colored segments and labelled with their squared distances ($\delta^2_{152,341} \approx 8{,}165$, $\delta^2_{152,345} \approx 8{,}073$ and $\delta^2_{152,272} \approx 7{,}901$). For each of these pairs, the squared Euclidean distance in the MDS map coincides with the corresponding entry of the RelMS squared distance matrix, so the objective of MDS is again satisfied.

Comparing the MDS done with the two different metrics, it can be seen that RelMS captures slightly less variability than Gower on the first two axes, but the resulting configuration is visually clearer. In the RelMS map, individuals form tight, well-separated clusters with noticeable gaps between groups and fewer in-between points. By contrast, the Gower map shows broader, more expanded clouds with greater within-group dispersion, which makes cluster boundaries less distinct.

```
disp(accum_r(1:10)')
16.8704
32.3625
44.9155
53.7592
62.5439
71.3088
80.0259
88.1350
93.5209
96.6501
```

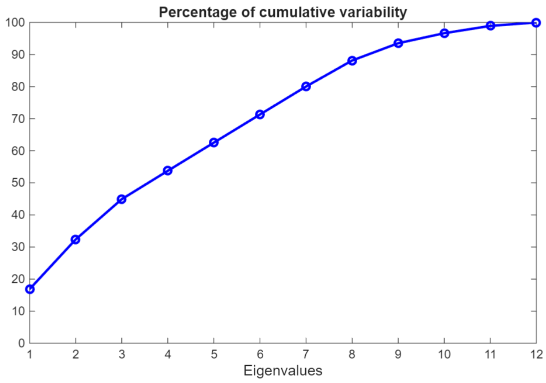


Figure. 1.4. Percentage of cumulative variability based on RelMS distance.

Furthermore, the plot in Figure 1.4 shows the cumulative percentage of total variability explained by the coordinates, computed from the eigenvalues of matrix **G** using RelMS. In comparison to the Gower analysis, using 9 coordinates explains 93.5 % of the variability, whereas with Gower reaching 90 % requires about 27 coordinates. This suggests that, under Gower, the dissimilarity information is distributed across many dimensions (a weaker low-dimensional structure), whereas RelMS produces a representation that is much more effectively summarized in a smaller number of coordinates.

1.2. Influence of the original variables on the MDS configuration

Since the original variables are available, it is of interest to examine how they influence the principal coordinates. To this end, we compute cross–correlation (or association) coefficients between the first three principal coordinates and the original variables, thereby assessing how strongly each variable contributes to each component. In this dataset, however, the number of binary and categorical variables is small compared to the quantitative variables. Therefore, for completeness, we still compute Cramér's V for the nominal variables (both binary and categorical).

This procedure is done for both the Gower's metric and the RelMS and displayed in Figure 1.5 and 1.6 respectively. We will start analyzing the correlation structure when Gower's metric is used:

- **First principal coordinate (PC1):** Although no single variable has an extremely high correlation with PC1, the strongest association is with the categorical variable

```
corr_table =

    0.0159    0.0233    0.0423
   -0.5333   -0.0267    0.0844
   -0.5546   -0.1744   -0.2492
   -0.3866   -0.0583   -0.0288
   -0.1616   -0.1393   -0.0029
   -0.3059   -0.0855    0.0345
    0.1140    0.9074    0.3660
    0.5091    0.4065    0.9606
    0.1564    0.1022    0.1279
    0.6894    0.1758    0.4724
```

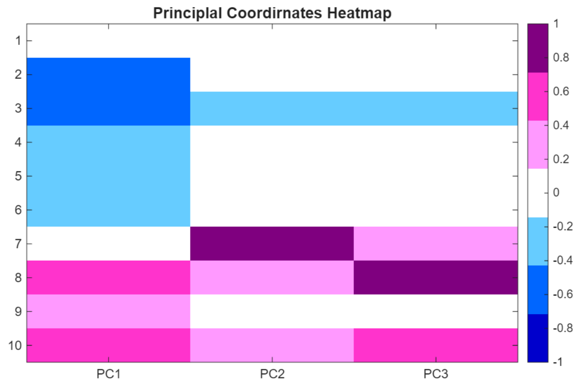


Figure. 1.5. Cross-correlation heatmap Gower distance.

type of ‘educational institution’ (X_{10}) (r = 0,6894). The next largest contributions come from the ‘expenditure’ variables ‘food’ (X_3) and ‘uniform’ (X_2), both negatively correlated with PC1 (r = −0,5546 and r = −0,5333, respectively), followed by the binary variable ‘help (X_8) (r = 0,5091).

- **Second principal coordinate (PC2):** The binary variable ‘sex’ (X_7) shows by far the strongest association with PC2 (r = 0,9074). Thus, PC2 can be interpreted as an almost direct sex axis. Since ‘sex’ is binary (e.g., coded as 0/1), a large positive correlation means that moving in the positive direction of PC2 corresponds to moving from the category coded as 0 toward the category coded as 1. In other words, the two sex groups tend to fall on opposite sides of PC2, so PC2 mainly separates individuals by sex.

- **Third principal coordinate (PC3):** PC3 is dominated by the binary variable ‘help’ (X_8), which has a strong positive correlation with this coordinate (r = 0,9606), with a smaller positive contribution from ‘type’ (X_{10}). This coordinate, therefore, separates individuals who receive help from those who do not.

In addition, when RelMS is used as the distance metric, the correlation structure exhibits the following patterns:

```
corr_table =

    0.0180    0.0471   -0.0295
    0.1361   -0.4627   -0.2017
    0.0090   -0.5820    0.2372
    0.0673   -0.3542   -0.0008
   -0.0850   -0.1267   -0.0107
    0.0032   -0.2564   -0.1456
    0.9074    0.4505    0.2541
    0.2620    0.5346    0.9670
    0.1057    0.1552    0.1036
    0.6236    0.6808    0.5071
```

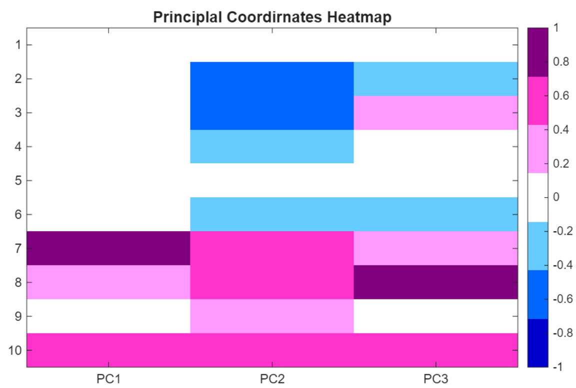


Figure. 1.6. Cross-correlation heatmap RelMS distance

- **First principal coordinate (PC1):** The binary variable 'sex' (X_7) shows the strongest association with PC1 (r = 0,9074), so the principal coordinate can be interpreted as an almost direct sex axis. In addition, 'type' (X_{10}) shows a moderate positive correlation, indicating a secondary influence of institution type on this coordinate.
- **Second principal coordinate (PC2):** Even though no variable shows an extremely high correlation with PC2, the strongest positive associations are with 'sex' (X_7) (r = 0,4505), 'help' (X_8) (r = 0,5346), and 'education' (X_{10}) (r = 0,6808). The largest negative correlations are with 'uniform' (X_2) (r = −0,4627) and 'food' (X_3) (r = −0,5820).
- **Third principal coordinate (PC3):** The binary variable 'help' (X_8) shows the strongest association with PC3 (r = 0,9670), so this coordinate can be interpreted as an almost direct help axis. In addition, 'type' (X_{10}) presents a moderate positive correlation (r = 0,5071), indicating a secondary influence of institution type on this coordinate.

The heatmaps in Figures 1.5 and 1.6 show that, under both Gower and RelMS distances, the MDS configuration is primarily driven by the binary variables. In particular, 'help' (X_8) is the most stable and dominant driver: it almost entirely defines the third coordinate in both metrics, so PC3 can be interpreted as an almost direct 'help' axis separating individuals who receive help from those who do not. Likewise, 'sex' (X_7) plays a central role in the geometry of the maps, although its contribution shifts across axes depending on the metric: under Gower it dominates PC2, whereas under RelMS it becomes the main driver of PC1. This indicates that separation by sex is a robust feature of the data, even if it is captured by different principal coordinates in the two solutions.

Beyond the binary effects, the main continuous drivers are the expenditure variables 'uniform' (X_2) and 'food' (X_3). Under Gower, their strongest contribution is concentrated on PC1, together with the categorical variable 'type' (X_{10}). Under RelMS, the influence of 'uniform' and 'food' is mainly expressed on PC2, where it is contrasted with a positive structural effect dominated by 'type' (X_{10}). Thus, both metrics point to the same influential variables, but they allocate their contributions differently across the leading coordinates. In the next section, we go beyond these global association measures by studying the partial influence of the most influential variables through Gower's interpolation formula, and we complement interpretation with conditional MDS scatterplots and a stability analysis of the resulting configuration.

1.3. Interpretation of the MDS configuration: Variable Contributions and Stability

The cross-correlation and association heatmaps in the previous section provide a global view of how the original variables relate to the leading principal coordinates: they indicate which variables are aligned with PC1–PC3 on average, but they do not explain how changes in a given variable affect the map locally, nor whether the observed geometry is robust to small perturbations of the sample. In this section, we address both issues for the RelMS-based MDS configuration by studying the variables' partial influence on the principal coordinates and by evaluating the stability of the configuration. Together, these analyses provide a clearer understanding of the factors that shape the MDS maps and of the reliability of the resulting representations. To examine these aspects, we use Gower's interpolation formula, which, given a Euclidean configuration computed on $n \in \mathbb{N}$ units, allows one to project an additional unit $n+1$ onto the existing map.

1.3.1. Partial influence of the variables on the principal coordinates

We begin by studying the partial influence of each original variable on the principal coordinates, in order to clarify how individual variables shape the MDS configuration. The goal is to identify how strongly each variable contributes to the positioning of the observations along the main dimensions and to clarify the role that individual variables play in shaping the overall structure of the map. By analyzing these partial effects, we gain a more detailed understanding of the relationships driving the configuration and the interpretability of the resulting dimensions.

The idea behind this procedure is to consider a single variable X_j, $j = 1, \dots, p$, and to project onto the MDS configuration the 'virtual' unit $v_t = (0, \dots, 0, t, 0, \dots, 0) \in \mathbb{R}^p$ for a wide range of values $t \in \mathbb{R}$, where the j-th entry is the only non-null component of v_t. For each t, we compute the dissimilarities between v_t and the n observed individuals using the same RelMS distance, and then project v_t onto the existing MDS configuration via Gower's interpolation formula. The resulting projected trajectory $y(v_t)_t$ provides a direct visualization of the influence of X_j: long, well-oriented trajectories indicate a strong geometric impact on the leading coordinates, whereas short or more diffuse trajectories suggest a weaker contribution.

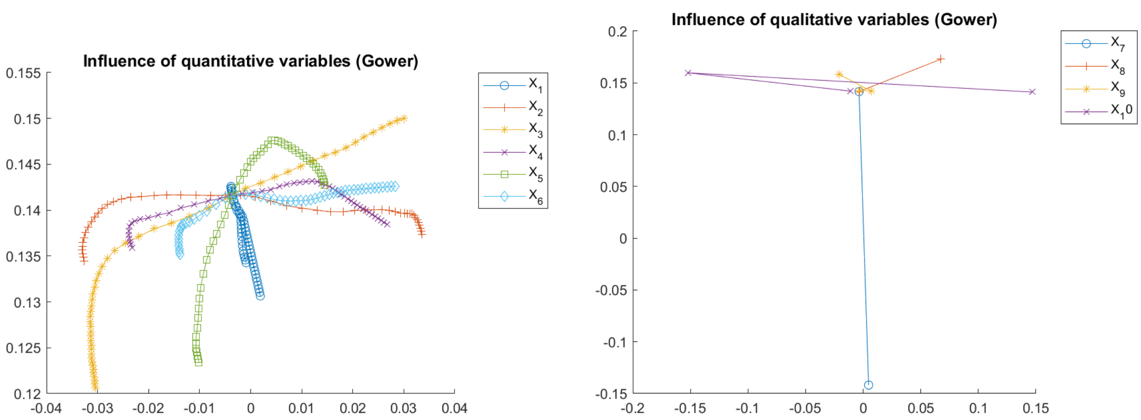


Figure. 1.7. Partial influence of the variables on the principal coordinates.

In Figure 1.7, each curve (quantitative variables) or segment (binary/categorical variables) represents the trajectory of the projected virtual units obtained by varying one variable while keeping the others fixed. Therefore, the spread of a trajectory along an axis measures how sensitive that principal coordinate is to that variable. For the quantitative variables (left panel), X_2 ('uniform'), X_4 ('books'), and X_6 ('activities') generate the widest horizontal displacements, indicating that they are more influential in the first principal coordinate. In contrast, X_1 ('age') and X_5 ('computer') show comparatively stronger vertical variation, indicating a larger influence on the second principal coordinate. On the other hand, for the qualitative variables (right panel), the length and direction of each segment summarize the separation between levels: X_{10} ('type') varies mostly along the horizontal direction and thus contributes primarily to PC1, whereas X_7 ('sex') produces the largest vertical separation, making it the dominant driver of PC2.

1.3.2. Profile identification: conditional scatterplots (MDS configuration obtained with RelMS)

Once the most influential variables have been identified, the next step is to visualize groups of individuals in the MDS maps. This can be easily done by producing conditional MDS maps, which are scatterplots of the principal coordinates conditioned to the values (for numeric variables) or to the categories (for binary and multiclass categorical variables) of the most influential variables. For simplicity, it is common to compute the two-dimensional conditional MDS maps, which correspond to the first two principal coordinates. It can also be insightful to analyze the conditional multiple scatter-plots of the first three principal coordinates.

Most influential continuous variables (Uniform and Food): The conditional MDS maps for the continuous expenditure variables shown in Figure 1.8, namely 'uniform' (X_2) and 'food' (X_3), suggest that these variables can help describe the groups observed in the configuration, but they do not appear to drive the global geometry. For 'uniform', the values are largely mixed within and across clusters, indicating substantial within-group variability and no clear gradient or separation in the map. In contrast, 'food' shows a slightly more structured behaviour: some clusters tend to concentrate higher or lower values, and a weak local gradient is visible across parts of the configuration, although the pattern is not strong enough to yield a clear-cut partition of individuals. Overall, both variables exhibit at most a moderate influence, refining the interpretation of clusters rather than explaining the main separations.

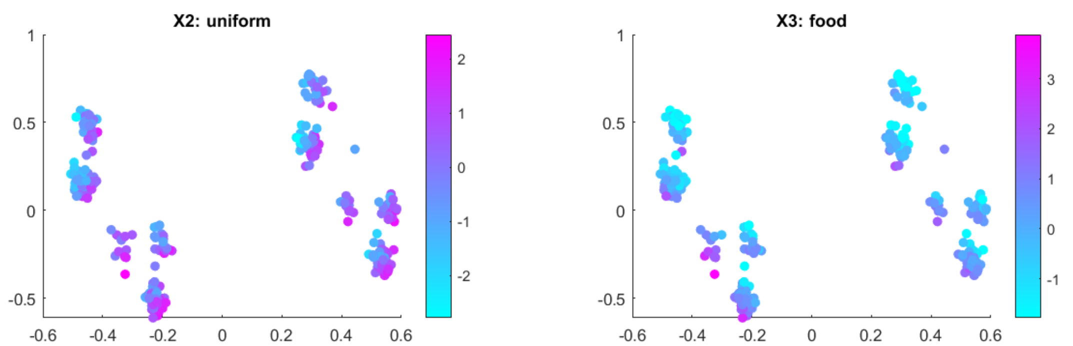


Figure. 1.8. Scatterplot of the first two Principal Coordinates conditioned by Uniform and Food.

Most influential binary variables (Sex and Help): In the scatterplot matrix conditioned on 'sex' (X_7) in Figure 1.9, the first principal coordinate clearly separates the two groups: males are mostly located on the right (positive PC1), whereas females concentrate on the left (negative PC1). On the other hand, in the multiple scatterplot conditioned by 'help', we can see that the third principal coordinate divides the individuals into two separate groups: the ones that do not receive any help (tend to be on the right, positive PC3) and those who do receive help (tend to be on the left, negative PC3).

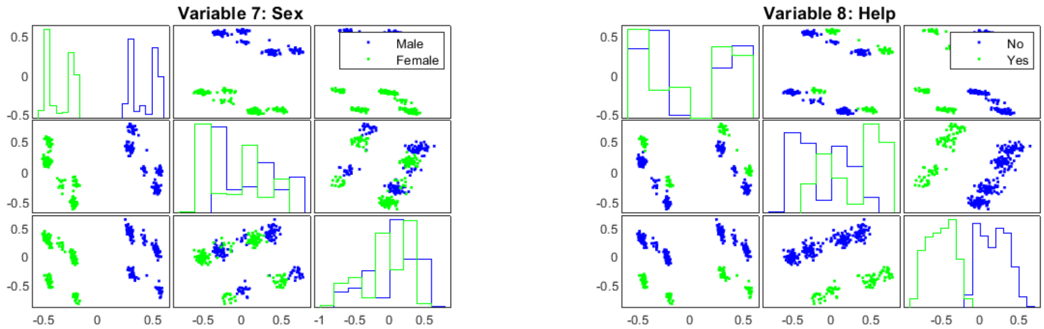


Figure. 1.9. Multiple Scatterplots of the PC1, PC2 and PC3 conditioned by Sex and Help.

Most influential categorical variables (Type): The three schooling types occupy partially overlapping but visibly differentiated regions of the first three principal coordinates, suggesting that school type is an important source of structural variation in the data. The clearest division can be seen in the second principal coordinate, where Public tends to be on the right (positive PC2), State-Funded Private on the left (negative PC2), and Private occupies a more central position.

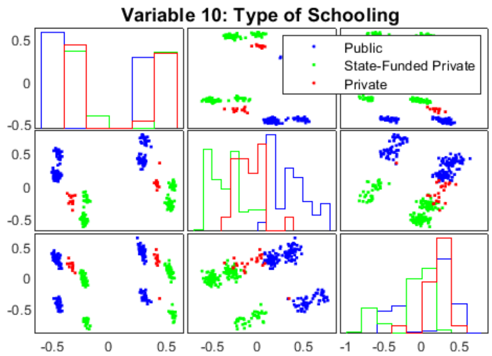


Figure. 1.10. Multiple Scatterplot of the first three Principal Coordinates conditioned by Type.

1.3.3. Stability of the MDS Configuration

Beyond interpretation, it is important to check how robust the MDS configuration is to small changes in the data, which requires additional information about the stability of the model. In this case, we are going to assess the stability of the MDS configuration by studying the influence of each unit on the MDS maps. For each $i = 1, ..., n$ we are going to delete the ith unit from the original dataset and compute the corresponding principal coordinates, which will be denoted by $\boldsymbol{Y}_{(i)}$. Then, Gower's interpolation formula will be used for projecting the *ith* unit onto the MDS configuration of $n-1$ points, obtaining the augmented configuration $\boldsymbol{Y}^*_{(i)}$. This procedure will be done in a crossvalidatory way, repeating the previous algorithm n times. Finally, confidence regions for the projections are obtained. For each unit i, we compute the Euclidean distances between the columns of Y_i and the columns of $Y_{(i)}$, and then we set the radius of the hypersphere equal to a certain quantile (90th or 95th) of the distribution of these distances. Thus, units with small hyperspheres around them are stable units, whereas large hyperspheres correspond to unstable units.

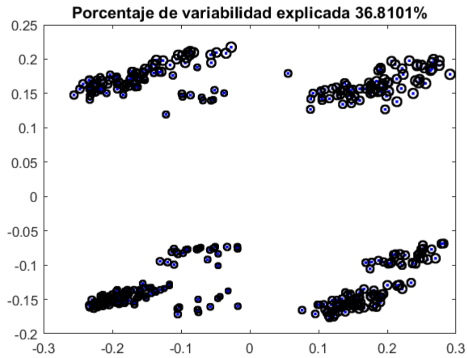


Figure. 1.11. Stability of the MDS Configuration.

It can be seen in Figure 1.11 that the confidence circles (hyperspheres in $\mathbb{R}^2$) around the points are generally small. The smallest circles correspond to the most stable units, whose projected positions have less variability when we perturb the data. Overall, this indicates that the MDS configuration obtained with Gower's distance is stable: the map and its main geometric structure remain essentially unchanged when small perturbations are applied to individual units.

2. CLUSTERING

Cluster analysis, also known as unsupervised classification or unsupervised pattern recognition, is a multivariate methodology that aims to discover homogeneous groupings within a dataset by exploiting the similarity structure between individuals. Formally, we consider a set $\mathcal{E}$ of n units together with a dissimilarity matrix $\mathbf{D} = (\delta_{ij})_{1\leq i,j\leq n}$, which defines the pairwise distances between these units: small values indicate very similar units, larger values indicate more distinct units, and $\delta_{ii} = 0$, where δ is a distance (or quasi-metric) defined in the previous chapter. The main objective is to identify clusters such that units within the same group are highly similar, while units belonging to different groups are dissimilar. This differs from discriminant analysis (supervised classification), since no class labels are provided; the groups are inferred from the data and, depending on the method, the number of clusters may be fixed in advance or selected from the data.

From a practical perspective, the utility of data clustering is generally observed in three main areas. First, it helps us understand the underlying structure of a dataset. Second, it establishes a natural classification based on the degree of similarity among individuals. Third, it functions as a compression technique by summarizing data through cluster prototypes, where a representative individual is used to characterize the properties of the entire cluster.

While the historical development of clustering methods is extensive, most modern approaches can be grouped into three broad families: hierarchical methods, partitional methods, and Bayesian (model-based) methods. In this chapter, we focus on hierarchical clustering and on representative partitional algorithms such as k-means and k-medoids(non-hierarchical), including practical aspects like selecting the number of clusters and interpreting clusters through visualization techniques.

2.1. Hierarchical Clustering

Hierarchical clustering organizes the individuals of the data into a nested sequence of groups. This nesting process can be done using agglomerative or divisive methods. On the one hand, agglomerative methods take a bottom-up approach, starting with individual units as groups and merging the two most similar groups on each iteration until all the subgroups are merged into a single cluster that contains all the individuals. On the other hand, divisive methods take a top-down approach, starting with the full set and splitting one group (the one with the highest internal dissimilarities) into two subgroups on each iteration. This process is repeated until each unit is a singleton cluster. In order to decide which clusters should be combined or where clusters should be split, a measure of dissimilarity between sets of units must be considered. This measure of dissimilarity will be known as the linkage criterion, and we will consider three types:

- **Single linkage:** the distance between two groups is the distance between their nearest members.
- **Complete linkage:** the distance between two groups is the distance between their farthest members.
- **Average linkage:** the distance between two groups is the average distance between their members.

Given a distance matrix $\boldsymbol{D}$, the results of any hierarchical clustering method that uses the linkage criterion α are displayed in a dendrogram, a tree-like diagram whose branches represent clusters. Then, there is a matrix of distances $\boldsymbol{U}_\alpha = (u_{\alpha,i,j})$ associated to the dendrogram that verifies the ultrametric property. A way of selecting a linkage criterion is to measure the degree of coherence between $\boldsymbol{D}$ and $\boldsymbol{U}_\alpha$, which can be done using the cophenetic correlation, defined as Pearson's correlation coefficient between the $\frac{n(n-1)}{2}$ pairs of distances $(\delta_{ij}, u_{\alpha,i,j})$, for $1 \leq i < j \leq n$. The idea is to choose the linkage criterion that yields the closest to one cophenetic correlation, as it is the one that 'deforms' the least the initial distance matrix $\boldsymbol{D}$.

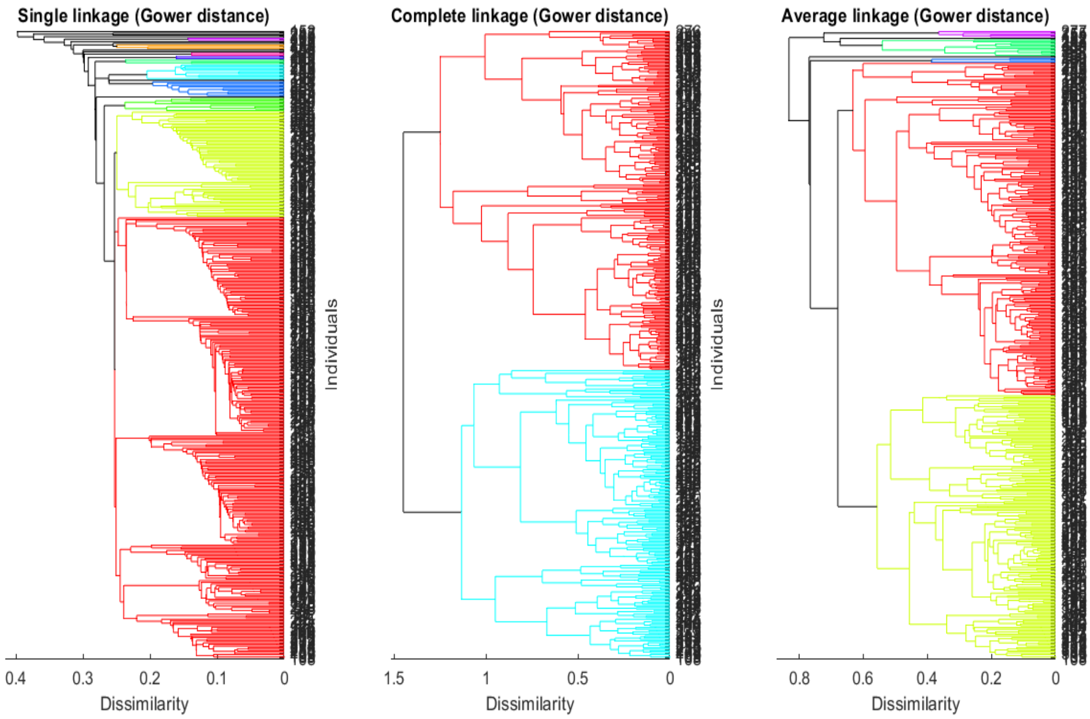


Figure. 2.1. Hierarchical clustering dendrograms.

The dendrograms obtained when performing hierarchical clustering with our dataset and considering Gower's distance matrix can be seen in Figure 2.1. The number of clusters in each of the dendrograms has been chosen visually by placing the threshold where the clusters are the farthest apart possible, or, in other words, where there is a big jump between two adjacent levels. The thresholds for the dendrograms using single, complete, and average linkage are 0.2528, 1.35, and 0.65, respectively, yielding the following 'best' numbers of clusters: 20, 2, and 7, respectively. Moreover, the cophenetic correlations of the dendograms are: 0.6788 (single linkage), 0.6596 (complete linkage), and 0.7474 (average linkage). Thus, since the average linkage criterion corresponds to the closest to 1 cophenetic correlation, it is the best choice for us and the best number of clusters is $k = 7$.

2.2. Non-hierarchical Clustering

2.2.1. K-means clustering

K-means clustering is an unsupervised classification method whose objective is to partition a set of individuals into k homogeneous groups, in such a way that observations within the same cluster are as similar as possible, while observations belonging to different clusters are as different as possible. Similarity between individuals is typically measured using the Euclidean distance, which is the default distance employed by the k-means algorithm.

Given a dataset of quantitative variables $\mathbf{X} \in \mathbb{R}^{n \times p}$, the k-means algorithm aims to minimize the within-cluster sum of squares (WCSS), defined as

$$\sum_{g=1}^{k} \sum_{i \in C_g} ||\mathbf{x}_i - \boldsymbol{\mu}_g||^2,$$

where C_g denotes the set of observations assigned to cluster g and $\boldsymbol{\mu}_g$ is the centroid (mean vector) of that cluster.

The algorithm proceeds iteratively. First, k initial centroids are selected. Then, each observation is assigned to the cluster whose centroid is closest in terms of Euclidean distance. After the assignment step, centroids are recomputed as the mean of the observations belonging to each cluster. These two steps are repeated until convergence, that is, until cluster assignments no longer change or the reduction in WCSS becomes negligible.

a) Choice of the number of clusters

A crucial aspect of k-means clustering is the selection of the number of clusters k. In this study, the number of clusters was chosen using the average silhouette coefficient. For each observation, the silhouette measures how similar it is to its own cluster compared to the nearest alternative cluster, taking values between -1 and 1. Higher values indicate a better clustering structure.

We evaluated solutions for $k = 2, \ldots, k_{\max}$, where $k_{\max}$ was set equal to the number of quantitative variables in the dataset. The optimal number of clusters was selected as the value of k that maximized the average silhouette coefficient. Based on this criterion, the optimal solution corresponds to $k = 2$ clusters.

b) k-means on raw quantitative data

First, k-means clustering was applied directly to the raw quantitative variables (after the preprocessing steps described in the previous project, including transformations and standardization). This approach groups individuals in the original variable space (taking into account the full-dimensional structure of the data), using Euclidean distances in the full p-dimensional space.

To assess clustering quality across candidate values of k, we examined two complementary criteria: the within-cluster sum of squares (WCSS) and the average silhouette coefficient. WCSS was inspected through the elbow pattern as k increases, while the silhouette coefficient evaluates clustering quality by comparing within-cluster cohesion with separation from the nearest neighbouring cluster, which served as the main guideline for selecting k.

Once the optimal number of clusters k was determined, the resulting groups were interpreted by analyzing the distribution of each quantitative variable within each cluster. This allowed us to characterize the clusters in terms of different expenditure patterns and demographic profiles.

c) k-means based on Principal Component Analysis

In this second approach, k-means clustering is not applied to the original quantitative variables, but to the matrix $\mathbf{Y}_q$ containing the scores of the first q principal components obtained from a Principal Component Analysis (PCA). In our study, we use $\mathbf{Y}_2$, i.e., the scores on the first two principal components.

More precisely, the PCA is computed using the function `comp2`, which transforms the preprocessed data matrix $\mathbf{Z}$ (standardized and transformed) into a new set of orthogonal variables. The matrix $\mathbf{Y}_2$ corresponds to the representation of the individuals in the space of principal components, where each column represents a principal component and each row corresponds to an individual. Formally, this transformation can be written as

$$\mathbf{Y} = \mathbf{Z}\mathbf{A},$$

where $\mathbf{A}$ is the matrix whose columns are the eigenvectors of the covariance (or correlation) matrix of $\mathbf{Z}$. As a result, the variables in $\mathbf{Y}_2$ are uncorrelated and ordered according to the amount of variability they explain.

The k-means algorithm is then applied to $\mathbf{Y}_2$, using the `MATLAB` function

$$\texttt{kmedias2}(\mathbf{Y}_2, k_{\max}),$$

meaning that distances between individuals are computed in the principal component space rather than in the original variable space. Consequently, clustering is driven by the main directions of variability in the data, while the effect of noise, correlation and scale differences among the original variables is reduced.

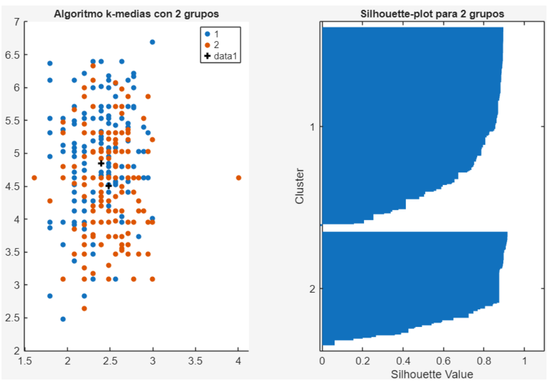
As in the clustering based on raw data, the number of clusters is selected using the average silhouette coefficient. This allows us to compare the clustering structure obtained from the original variables with the one obtained after the PCA transformation.

d) Comparison of k-means clustering on raw data and PCA

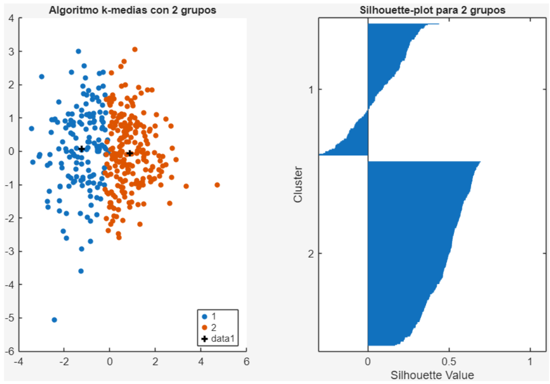
Figure 2.2 compares the results of the k-means algorithm with $k = 2$ clusters when applied to the raw quantitative variables and when applied to the PCA representation of the data.

From a visual perspective, when k-means is applied directly to the raw quantitative variables, the separation between clusters is less apparent in the graphical representation. The clusters exhibit a higher degree of overlap, making the visual identification of the groups less straightforward.

In contrast, the clustering based on the principal component scores shows two clearly distinguishable clouds of points when individuals are represented in the space of the first principal components. This visual separation suggests that the main directions of variability captured by the PCA reveal an underlying grouping structure in the data, which becomes more evident in the reduced-dimensional representation.



(a) k-means clustering on raw quantitative data



(b) k-means clustering on PCA scores

Figure. 2.2. Comparison of k-means clustering results for $k = 2$ clusters using raw data and PCA-based representation.

Despite this visual difference, the average silhouette coefficient is higher for the clustering based on the raw data ($\bar{s} = 0{,}7515$) than for the PCA-based clustering ($\bar{s} = 0{,}3118$). This indicates that, in the original variable space, observations are on average more clearly separated from the alternative cluster in terms of Euclidean distance.

However, the silhouette coefficient is a local, distance-based measure and does not fully capture the global multivariate structure of the data. In this context, the PCA-based clustering is preferred because it is performed in an orthogonal space that concentrates the main sources of variability, removes linear correlations among variables, and reduces the effect of redundant information.

Therefore, although the silhouette value is lower, the k-means solution obtained from the PCA scores is retained for further analysis, as it provides a more stable and interpretable representation of the clustering structure.

e) Interpretation of the clusters

To interpret the final clustering solution, the mean values of the original quantitative variables ('age', 'uniform', 'food','books', 'computer' and 'activities') were computed separately for each cluster. Table 2.1 reports the mean values of the quantitative variables by cluster.

The results show that the two clusters have very similar average age values, indicating that age does not play a relevant role in differentiating the groups. In contrast, clear differences are observed across all expenditure-related variables.

Table 2.1. Mean values of the quantitative variables by cluster.

Variable	Cluster 1	Cluster 2
Age	2.43	2.42
Uniform	5.17	4.09
Food	1131.4	630.8
Books	5.27	4.18
Computer	5.53	4.84
Activities	5.98	4.76

The first cluster is characterised by systematically higher average expenditure across all spending categories. In particular, individuals in this group present higher mean values in food-related expenses, as well as in books, computer equipment and extracurricular activities. This pattern reflects a higher overall level of educational spending.

In contrast, the second cluster exhibits lower average values for the same expenditure variables, indicating a more moderate educational expenditure profile. The consistency of lower spending across multiple categories suggests that this group is associated with a more restrained level of investment in educational resources.

Taken together, these results indicate that the clustering primarily differentiates individuals according to their overall level of educational expenditure, rather than demographic characteristics such as age.

2.2.2. k-medoids clustering

K-medoids clustering is an unsupervised classification method closely related to k-means, but with a key methodological difference. While k-means represents each cluster by its centroid (the mean of the observations in the group), k-medoids represents each cluster by a medoid, which is an actual observation from the dataset that minimises the total distance to the other observations in the same cluster.

Given a dataset of quantitative variables $X \in \mathbb{R}^{n\times p}$, the k-medoids algorithm aims to partition the individuals into k clusters by minimising the sum of distances between each observation and the medoid of its assigned cluster. Unlike k-means, which relies on squared Euclidean distances, k-medoids can be defined using more general distance measures.

A major advantage of k-medoids is its robustness to outliers. Since medoids are real observations rather than mean vectors, extreme values have a reduced influence on the clustering solution. This property makes k-medoids particularly suitable for datasets with skewed distributions or the presence of atypical observations.

The algorithm proceeds iteratively by selecting an initial set of medoids, assigning each observation to the nearest medoid according to the chosen distance, and then updating the medoids to improve the clustering objective. These steps are repeated until convergence is reached.

Visualization in the original variable space

We apply the k-medoids clustering algorithm to the quantitative variables of the dataset. Unlike k-means, which represents each cluster by its centroid, k-medoids represents each cluster by a medoid, that is, an actual observation of the dataset that minimizes the sum of distances to the other observations in the same cluster. This makes the method more robust to outliers and extreme observations.

Since the quantitative variables are measured on different scales and exhibit different levels of variability, the data are first standardized. The clustering is then performed using the Mahalanobis distance, which accounts for the covariance structure of the variables and is therefore more appropriate than the Euclidean distance in a multivariate setting. The number of clusters is fixed to $k = 2$.

Figure 2.3 displays the results of the k-medoids clustering using pairwise scatterplots of the original variables. Observations are coloured according to their cluster membership, while the medoids are highlighted.

From these plots, we observe that the two clusters partially overlap when projected onto pairs of original variables. In particular, no single pair of variables provides a clear linear separation between the clusters. This behavior is expected, since the clustering is performed in the full multivariate space and the figures only represent two-dimensional projections of a higher-dimensional structure. Nevertheless, differences in location and dispersion between the two groups can be observed, especially in projections involving the first variable, indicating that the clusters differ in their overall multivariate profiles.

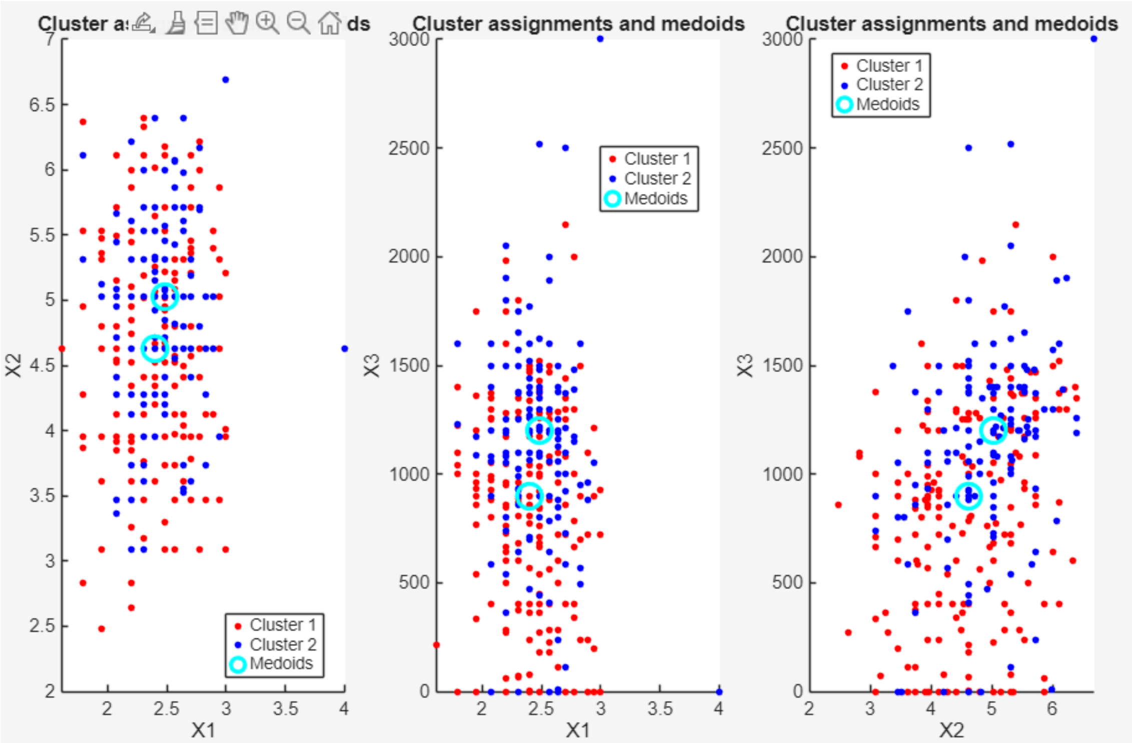


Figure. 2.3. Pairwise scatterplots of the original variables with *k*-medoids clustering results.

Multidimensional scaling (MDS) representation

To obtain a low-dimensional representation that preserves the multivariate distance structure used by the clustering algorithm, we apply classical multidimensional scaling (MDS), also known as Principal Coordinates Analysis (PCoA), to the Mahalanobis distance matrix.

The resulting MDS coordinates define a new orthogonal system of axes, ordered according to the amount of distance variability they explain. Figure 2.4 shows the *k*-medoids clustering results in this MDS space, including the projected medoids.

In contrast to the raw-variable projections, the MDS representation reveals two clearly differentiated clouds of points, corresponding to the two clusters identified by the *k*-medoids algorithm. The separation between the groups is more evident in this space, reflecting the fact that MDS preserves the same Mahalanobis distance structure used in the clustering procedure.

Furthermore, the medoids are located near the centres of their respective point clouds, confirming their role as representative observations of each cluster and indicating good within-cluster cohesion. This clearer separation in the MDS space supports the validity of the two-cluster solution obtained by the *k*-medoids algorithm.

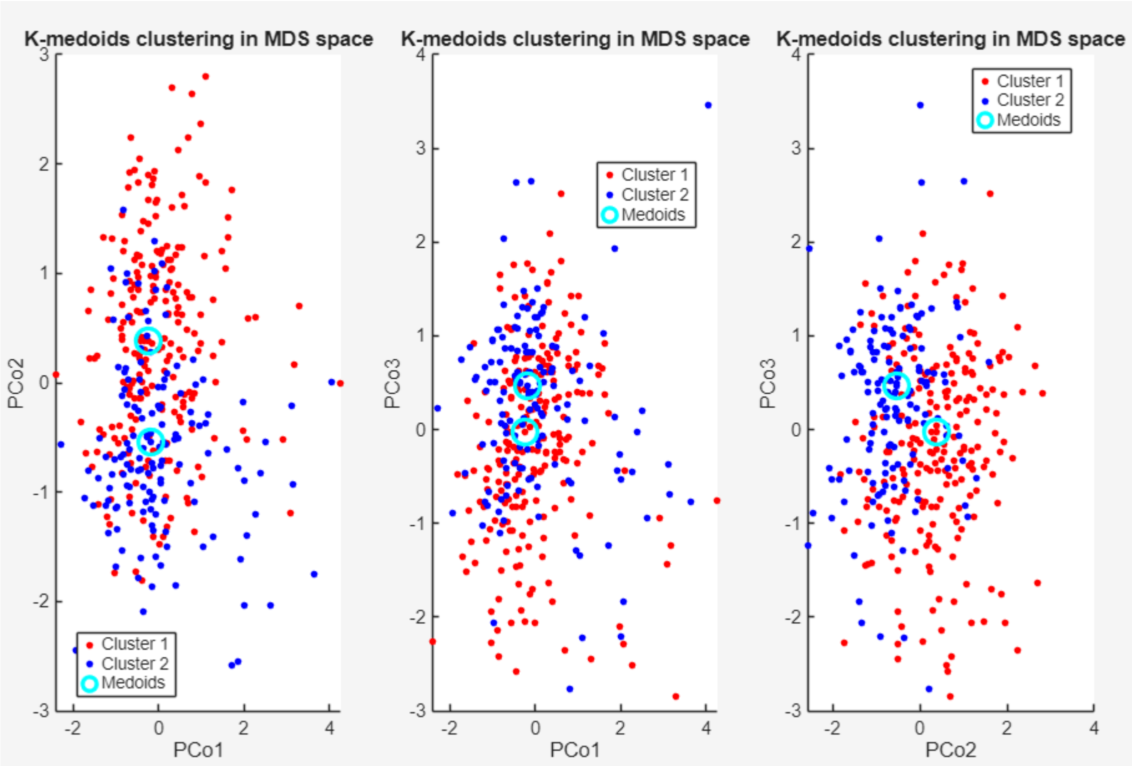


Figure. 2.4. MDS representation based on the Mahalanobis distance matrix.

Overall, while the raw-data scatterplots provide an interpretable view in the original measurement units, the MDS representation offers a geometrically faithful visualisation of the clustering structure and allows for a clearer assessment of cluster separation.

Conclusions

In this work, we analyzed household educational expenditure data using a combination of distance-based methods, dimension reduction techniques and clustering algorithms. Multidimensional Scaling (MDS) proved to be a suitable exploratory tool for mixed-type data, allowing us to visualize the structure of the dataset based on Gower and RelMS distances. The RelMS metric provided a clearer and more stable low-dimensional representation, with better-defined group structures.

Clustering analyses, including hierarchical methods, k-means and k-medoids, revealed the presence of meaningful groupings in the data. In particular, the k-means solution identified two clearly differentiated clusters, mainly separating individuals with higher overall educational expenditure from those with lower expenditure levels, while demographic variables played a secondary role.

Overall, the combination of MDS and robust clustering techniques offers an effective and interpretable framework for uncovering structure in complex, mixed-type datasets.